



MMM 5162 – MODELLING AND SIMULATION

Fall Semester 2026 – Term 481

Mini Project 1

[CLO 3.1 | PLO V2]

Computational Simulation and Parameter Study

**Submitted To
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1. Purpose

This project simulates one smooth pick-and-place cycle of a two-link planar robot arm. Both joints follow a cubic (smoothstep) motion law that guarantees zero angular velocity at the start and end of the motion, avoiding any velocity discontinuity.

The goals are:

1. implement the model as reusable functions,
2. study how the nominal cycle time T affects the required joint speed
3. extract quantitative performance measures from the simulation.

2. Model and Parameters

The robot's motion is determined using standard time and applied in the following kinematics equations:

$$u = t/T, s(u) = 3u^2 - 2u^3, q_j(t) = q_{j,start} + (q_{j,end} - q_{j,start})s(u), j = 1,2$$

$$x = L1\cos(q1) + L2\cos(q1 + q2), y = L1\sin(q1) + L2\sin(q1 + q2)$$

And parameters value we have for robot are:

Table 1 Assigned parameters for the two-link planar robot model

Parameter	Value	Parameter	Value
L1 (m)	0.420	q2,start (deg)	-48.6
L2 (m)	0.332	q2,end (deg)	19.0
q1,start (deg)	28.0	Nominal T (s)	2.72
q1,end (deg)	71.0		

3. Results and Analysis (Baseline, T = 2.72 s)

For the baseline cycle time both joints move continuously from their initial to final positions. The resulting end-effector trajectory is smooth and monotonic, with no visible reversal or discontinuity, confirming that the implemented cubic motion produces the intended smooth pick-and-place motion, as shown in Figure 1.

The baseline quantitative results are summarized in Table 2. Joint 1 moves through 43.0°, while Joint 2 moves through a larger angular displacement of 67.6°. Consequently, Joint 2 reaches the higher maximum absolute angular velocity of 0.65 rad/s, compared with

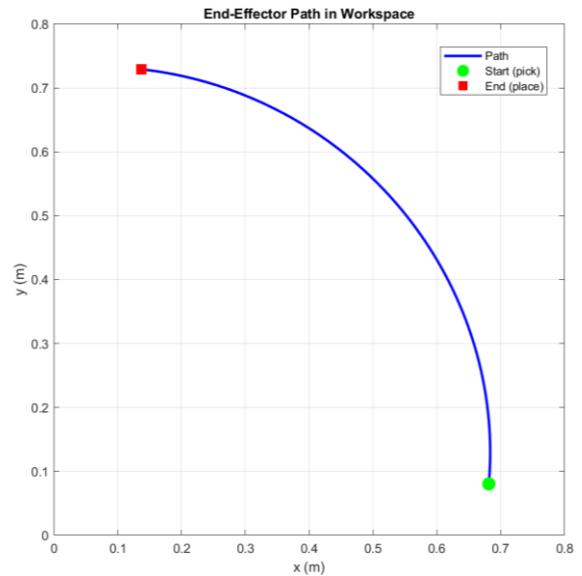


Figure 1 End-effector path in the workspace, from the pick point to the place point

0.41 rad/s for Joint 1. The corresponding velocity profiles are shown in Figure 2.

Table 2 Baseline simulation results for the nominal cycle time $T=2.72s$

Metric	Value
Max $ q1dot $ (rad/s)	0.4138
Max $ q2dot $ (rad/s)	0.6506
Path length (m)	0.9370
x_start (m)	0.6816
y_start (m)	0.0804
x_end (m)	0.1367
y_end (m)	0.7291

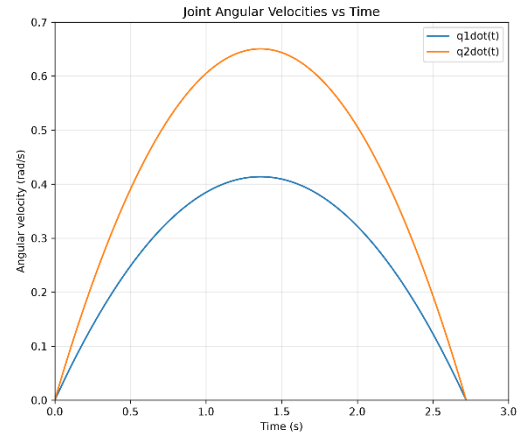


Figure 2 Joint angular velocity profiles for Joint 1 and Joint 2 during the baseline cycle.

4. Comparing - Parameter Study (0.8T, T, 1.2T)

The effect of cycle time on joint speed was evaluated using the required (0.8T), (T), and (1.2T) cases. The results are shown below and compared in Figure 3 and Table 3.

The results show a clear inverse relationship between cycle time and the required joint angular velocity. Reducing the cycle time from (T) to (0.8T) increases the maximum velocity by approximately 25% for both joints. In contrast, increasing the cycle time to (1.2T) reduces the maximum velocity by approximately 16.7%.

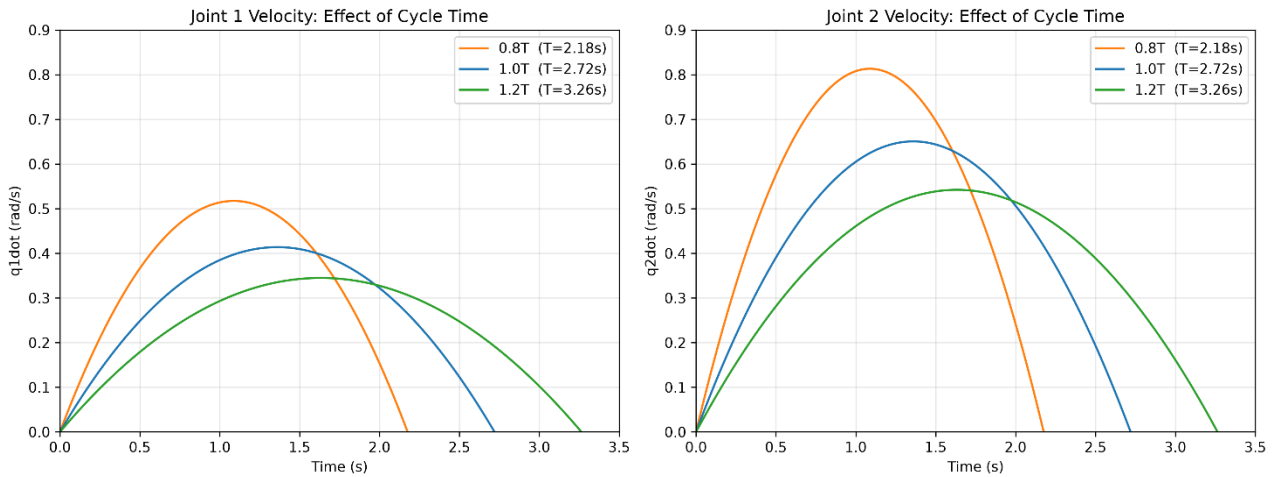


Figure 3 Effect of cycle time on the maximum angular velocity of Joint 1 and Joint 2.

Table 3 Effect of cycle time on the maximum joint angular velocities

Case	T_s	max_q1dot_rad_s	max_q2dot_rad_s
0.8T	2.176	0.5173	0.8133
1.0T	2.72	0.4139	0.6506
1.2T	3.264	0.3449	0.5422

Appendix

The code and data results found in the GitHub repository:



<https://github.com/Alidev29/MMM-5162--MODELLING-AND-SIMULATION/tree/main/MINI-PROJECTS/MP1%20-%20Robot%20Model%2C%20Workspace%20and%20Drive%20Feasibility>